

NGUYEN KHAC BAO

AI Engineer Intern – Algorithm & Agent

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CAREER OBJECTIVE

AI Engineer Intern candidate focused on building practical AI systems combining LLM/RAG agents, backend APIs, computer vision pipelines, and deployable inference services. Hands-on with Python, FastAPI, Gemini API, YOLOv8n, OpenCV, Docker, Cloud Run, and real-time AI demos.

EDUCATION

Posts and Telecommunications Institute of Technology (PTIT)

2022 – 2027

Bachelor of Information Technology

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision.

TECHNICAL SKILLS

AI / LLM / Agent: RAG pipelines, Gemini API, embeddings, vector search, prompt engineering, chatbot workflow

Backend / API: Python, FastAPI, REST API, WebSocket, Streamlit, JSON/API integration

ML / Computer Vision: PyTorch, YOLOv8n, OpenCV, OCR, object detection, tracking, dataset QA

Cloud / DevOps: Docker, Linux, Git/GitHub, Google Cloud Run, Cloud SQL, Cloud Build, Vast.ai

Data: SQL basics, CSV/JSON processing, metadata tracking, evaluation reports

LANGUAGES

Vietnamese: Native | English: TOEIC 550/990; technical reading proficiency for documentation, model cards, API references, and engineering articles.

PROJECTS

Vietnamese Legal RAG Chatbot

2026

Personal Project | Python, FastAPI, Streamlit, Gemini API, Qdrant Cloud, Cloud SQL, Docker, Google Cloud Run | Source: [GitHub](#)

- Built a cloud-ready RAG chatbot with FastAPI backend and Streamlit frontend for retrieval-grounded Vietnamese legal question answering.
- Integrated Gemini API, embeddings, and Qdrant vector search to retrieve relevant context before generating answers.
- Stored chat history with Cloud SQL/MySQL and configured API/database credentials through environment-based deployment.
- Designed extension points for async document indexing, OpenAI-compatible LLM serving, and future AI agent workflows.

Real-Time Face Mask Compliance Detection System

2026

Personal Project | Python, FastAPI, WebSocket, YOLOv8n, OpenCV, Docker, Google Cloud Run | Source: [GitHub](#)

- Trained a YOLOv8n 3-class mask compliance detector: correct_mask / incorrect_mask / no_mask, achieving 0.970 mAP@50 and 0.749 mAP@50:95 on validation.
- Built a reproducible dataset pipeline to convert VOC/XML annotations to YOLO format, audit class distribution, and track edge cases such as lighting, occlusion, blur, and reflection.
- Developed a FastAPI REST + WebSocket inference service with browser webcam UI for real-time detection, violation counting, and test-frame capture.
- Containerized and deployed the service to Google Cloud Run; smoke-tested public REST/WSS inference with warm latency around 151 ms REST and 127 ms WSS.

Traffic Violation Detection & License Plate Recognition

2026

Personal Project | Python, OpenCV, Ultralytics YOLO, ByteTrack, HyperLPR3, JSON | Source: [GitHub](#)

- Built an end-to-end fixed-camera CV pipeline for vehicle detection, ByteTrack tracking, traffic-light ROI state, stop-line crossing logic, plate OCR, and JSON evidence generation.
- Fine-tuned a YOLO plate detector on 8,598 CCPD images, achieving 0.989 precision, 0.998 recall, and 0.994 mAP50 on validation.
- Designed a reproducible license plate dataset workflow with annotation parsing, YOLO bbox conversion, OCR crops, negative samples, train/val/test split, and QA reports.
- Analyzed real-world issues such as low-light images, heavy blur, small/tilted plates, weather conditions, hard negatives, and missing/invalid YOLO boxes.

AutoFall Labeler Tool

2026

Additional Project | Python, Streamlit, OpenCV, YOLO-Pose, Pandas | Source: [GitHub](#)

- Built a human-in-the-loop dataset assistant for fall-detection data with video frame extraction, YOLO-Pose suggestions, Streamlit review UI, QA reports, and reviewed-only export.
- Designed CSV-based source-of-truth records for metadata, annotations, keypoints, and review events to support reproducible dataset versioning.
- Reviewed and cleaned a 656-frame CCTV benchmark, producing 332 reviewed exportable samples with 90.2% coarse fall/non-fall agreement.

ACTIVITIES

AI Research Group – University Club

2025 – Present

Member

Discussed Machine Learning, Deep Learning, Computer Vision, RAG/LLM applications, and AI system development.